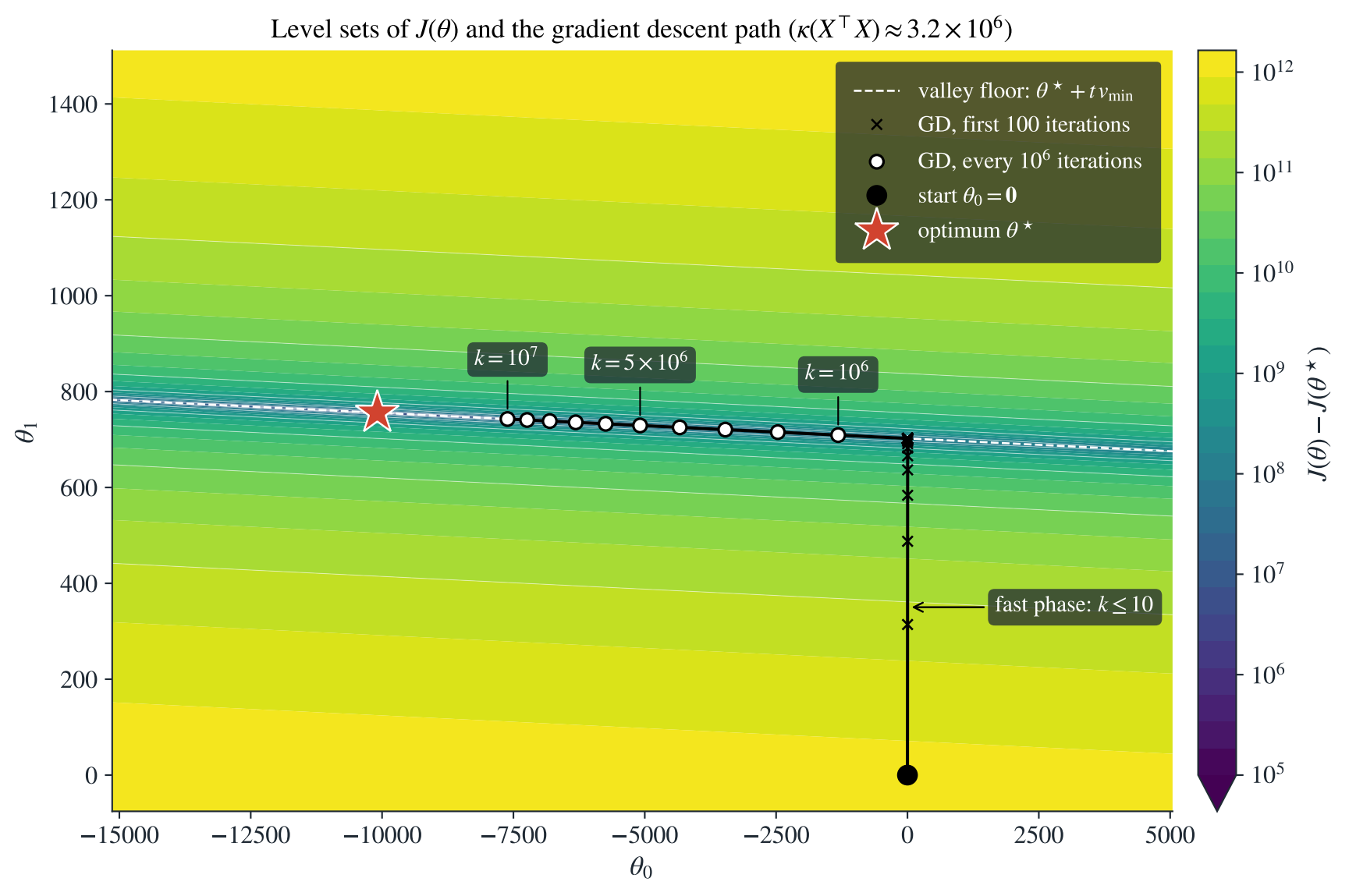

Level sets of $J(\theta)$ and the gradient descent path ($\kappa(X^\top X) \approx 3.2 \times 10^6$)
valley floor: $\theta^\star + t v_{\min}$
GD, first 100 iterations
GD, every 10^6 iterations
start $\theta_0 = \mathbf{0}$
optimum $\theta^\star$
$k = 10^7$
$k = 5 \times 10^6$
$k = 10^6$
fast phase: $k \leq 10$
θ_1
θ_0
$J(\theta) - J(\theta^\star)$
1400
1200
1000
800
600
400
200
0
−15000
−12500
−10000
−7500
−5000
−2500
0
2500
5000
10^{12}
10^{11}
10^{10}
10^9
10^8
10^7
10^6
10^5